

Question

A complex oxide ABO_2 can form multiple crystal polymorphs. In the crystal structures involved in this problem, the same type of metal ions occupy identical interstitial sites, with the same coordination number and coordination polyhedra.

For the cubic phase $\alpha - \text{ABO}_2$, X-ray diffraction measurements revealed several characteristic interplanar spacings of the crystal as 239.9, 207.7, 146.9, 125.3, and 103.9 pm. The density of this crystal is known to be $4.389 \text{ g} \cdot \text{cm}^{-3}$ with an error margin below 0.08%.

For $\gamma - \text{ABO}_2$, the unit cell parameters were measured as $a = b = 405.7 \text{ pm}$, $c = 875.9 \text{ pm}$, with γ unknown. This crystal can be described as: A and B alternately filling interstitial sites within layers to form polyhedral layers, where all polyhedral layers share identical chemical environments.

For another ABO_2 structure, the unit cell parameters were measured as $a = b = 296.3 \text{ pm}$, $c = 1463.6 \text{ pm}$, with γ unknown. This crystal can be viewed as having A and B arranged in alternating layers, where atoms of the same type share identical chemical environments.

Determine the identities of A and B before answering.

- A.** ABO_2 dissolves in hydrochloric acid under certain conditions, resulting in a light pink solution.
- B.** $\gamma - \text{ABO}_2$ has the same crystal structure as chalcopyrite.
- C.** $\alpha - \text{ABO}_2$ belongs to the simple cubic lattice.
- D.** From the ratio of the unit cell volumes of different crystal forms, it can be concluded that both $\gamma - \text{ABO}_2$ and the third ABO_2 belong to the tetragonal crystal system.
- E.** Based on the unit cell volume and lattice parameters, in both $\alpha - \text{ABO}_2$ and $\gamma - \text{ABO}_2$, the O atoms adopt either exact or approximate face-centered cubic close packing.

F. At the same temperature and pressure, the standard entropy of $\alpha - \text{ABO}_2$ is lower than that of $\gamma - \text{ABO}_2$.

G. All other options are incorrect

Answer

Correct Answer: **E**

Detailed Explanation

Analysis of the crystal structure of $\alpha - \text{ABO}_2$ and elements A, B

$\alpha - \text{ABO}_2$ belongs to the cubic crystal system, with given interplanar spacing d values: 239.9 pm, 207.7 pm, 146.9 pm, 125.3 pm, and 103.9 pm.

For the cubic crystal system, the relationship between interplanar spacing and lattice parameter a is:

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2}$$

Based on the proportions, the d values of the interplanar spacings correspond to $a/\sqrt{3}$, $a/\sqrt{4}$, $a/\sqrt{8}$, $a/\sqrt{11}$, and $a/\sqrt{16}$.

The lattice parameter is determined as $a = 415.5$ pm.

CHECKPOINT

1 PTS

$$a_{\alpha} = 415.5 \text{ pm}$$

Moreover, the observed $h^2 + k^2 + l^2$ values follow the rules of a face-centered cubic (FCC) lattice (i.e., no extinction when h , k , and l are all odd or all even). Thus, the α phase adopts a face-centered cubic lattice.

CHECKPOINT

1 PTS

The α phase has a face-centered cubic lattice

From the density formula:

$$\rho = \frac{ZM}{N_A a^3}$$

By enumerating the possible M values for different Z , the smallest error occurs when $Z = 2$, with A and B being Li and Fe, respectively.

CHECKPOINT

3 PTS

A, B = Li, Fe

Given that $a_\gamma \approx a_\alpha$ and $c_\gamma \approx 2a_\alpha$, the γ phase can be considered as a combination of two cubic unit cells. Therefore, the γ phase adopts a tetragonal unit cell.

CHECKPOINT

1 PTS

γ – LiFeO_2 belongs to the tetragonal crystal system

The reduction in symmetry indicates that the occupancy of Li and Fe sites transitions from disordered to ordered.

CHECKPOINT

2 PTS

$\alpha - \text{LiFeO}_2$ is a disordered structure, while $\gamma - \text{LiFeO}_2$ is an ordered structure

For the other structure, $a \approx \frac{1}{\sqrt{2}}a_\alpha$ and $c \approx 2\sqrt{3}a_\alpha$. The proportionality coefficients suggest that this structure adopts a hexagonal unit cell, with one O atom per layer and a total of six layers. It likely corresponds to a face-centered cubic close-packed arrangement, where Li and Fe occupy octahedral voids in different layers. The positional arrangement can be represented as $Ac_{\text{Li}}Ba_{\text{Fe}}Cb_{\text{Li}}Ac_{\text{Fe}}Ba_{\text{Li}}Cb_{\text{Fe}}A$.

CHECKPOINT

2 PTS

The third layered structure belongs to the trigonal crystal system with an hR lattice

Analysis of the options:

- A. FeCl_3 is yellow, not light pink;
- B. LiFeO_2 fills octahedral voids, whereas chalcopyrite CuFeS_2 fills tetrahedral voids;
- C. $\alpha - \text{ABO}_2$ adopts a face-centered cubic lattice;
- D. The third layered ABO_2 belongs to the trigonal crystal system;
- E. Correct;
- F. $\alpha - \text{ABO}_2$ is a disordered phase, while $\gamma - \text{ABO}_2$ is an ordered phase. Under the same conditions, $\alpha - \text{ABO}_2$ has higher entropy.